Parity and invariants

High school · reasoning · Synthesis problem · 01

Problem · Investigate, justify, conclude. An 8×8 chessboard is coloured alternately black and white. Remove two diagonally opposite corners. Can the 62 remaining squares be tiled by 31 dominoes of size 1×2, each covering two edge-adjacent squares? Count the removed colours and those covered by a domino. Compare with removing two neighbouring corners.